

## Tutorial 10 Question

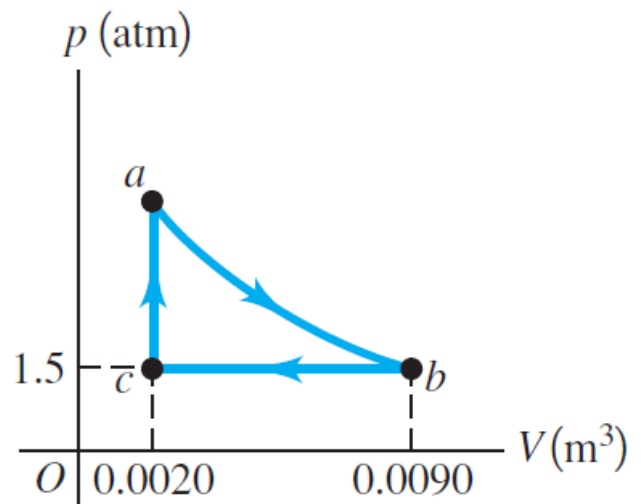
### Heat Engine, Refrigerator & Heat Pump

#### Question 1      Gasoline Engine

**20.3 • A Gasoline Engine.** A gasoline engine takes in  $1.61 \times 10^4$  J of heat and delivers 3700 J of work per cycle. The heat is obtained by burning gasoline with a heat of combustion of  $4.60 \times 10^4$  J/g. (a) What is the thermal efficiency? (b) How much heat is discarded in each cycle? (c) What mass of fuel is burned in each cycle? (d) If the engine goes through 60.0 cycles per second, what is its power output in kilowatts? In horsepower?

#### Question 2      $pV$ -diagram of a heat engine

**20.5 ••** The  $pV$ -diagram in Fig. E20.5 shows a cycle of a heat engine that uses 0.250 mole of an ideal gas having  $\gamma = 1.40$ . The curved part  $ab$



of the cycle is adiabatic. (a) Find the pressure of the gas at point  $a$ . (b) How much heat enters this gas per cycle, and where does it happen? (c) How much heat leaves this gas in a cycle, and where does it occur? (d) How much work does this engine do in a cycle? (e) What is the thermal efficiency of the engine?

### Question 3      Mercedes-Benz SLK320 vs Dodge Viper GT2



The Mercedes-Benz SLK230 is a discontinued model, so there are no new ones for sale; you can only purchase a used one. Used prices vary significantly based on condition, mileage, and location, but the average sale price is around \$8,928, with recent sales ranging from as low as \$3,000 to as high as \$20,500.

Dodge Viper GT2  
USD\$147,965.00

**20.7 ••** The Otto-cycle engine in a Mercedes-Benz SLK230 has a compression ratio of 8.8. (a) What is the ideal efficiency of the engine? Use  $\gamma = 1.40$ . (b) The engine in a Dodge Viper GT2 has a slightly higher compression ratio of 9.6. How much increase in the ideal efficiency results from this increase in the compression ratio?

### Question 4      Different units for refrigerator

**20.8 •** The coefficient of performance  $K = H/P$  is a dimensionless quantity. Its value is independent of the units used for  $H$  and  $P$ , as long as the same units, such as watts, are used for both quantities. However, it is common practice to express  $H$  in Btu/h and  $P$  in watts. When these mixed units are used, the ratio  $H/P$  is called the energy efficiency rating (EER). If a room air conditioner has a coefficient of performance  $K = 3.0$ , what is its EER?

### Question 5      Time for Cooling

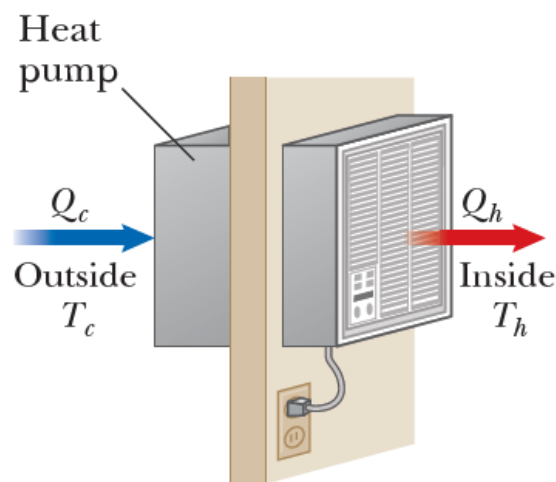
**20.11 ••** A refrigerator has a coefficient of performance of 2.25, runs on an input of 95 W of electrical power, and keeps its inside compartment at 5°C. If you put a dozen 1.0-L plastic bottles of water at 31°C into this refrigerator, how long will it take for them to be cooled down to 5°C? (Ignore any heat that leaves the plastic.)

### Question 6      Heat transfer of a refrigerator

**20.12 ••** A freezer has a coefficient of performance of 2.40. The freezer is to convert 1.80 kg of water at  $25.0^{\circ}\text{C}$  to 1.80 kg of ice at  $-5.0^{\circ}\text{C}$  in one hour. (a) What amount of heat must be removed from the water at  $25.0^{\circ}\text{C}$  to convert it to ice at  $-5.0^{\circ}\text{C}$ ? (b) How much electrical energy is consumed by the freezer during this hour? (c) How much wasted heat is delivered to the room in which the freezer sits?

### Question 7      Heat Pump

17. A heat pump used for heating shown in Figure P21.17 is essentially an air conditioner installed backward. It extracts energy from colder air outside and deposits it in a warmer room. Suppose the ratio of the actual energy entering the room to the work done by the device's motor is 10.0% of the theoretical maximum ratio. Determine the energy entering the room per joule of work done by the motor given that the inside temperature is  $20.0^{\circ}\text{C}$  and the outside temperature is  $-5.00^{\circ}\text{C}$ .

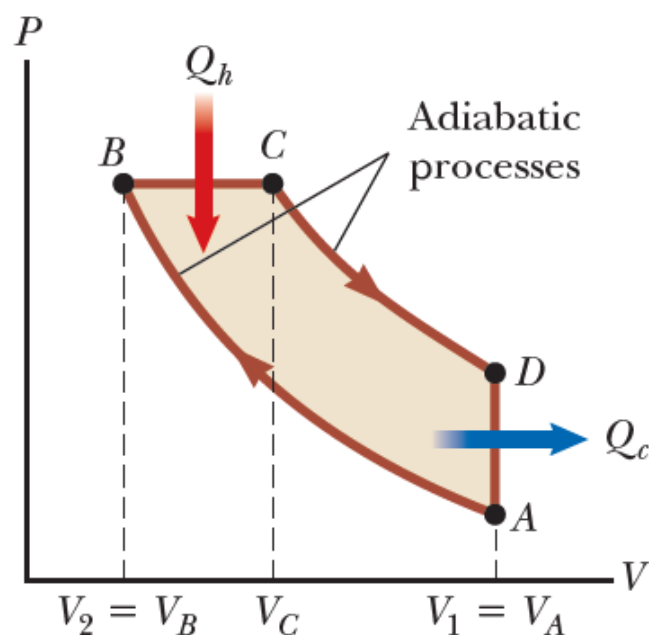


**Figure P21.17**

### Question 8      An idealized diesel engine

- 19.** An idealized diesel engine operates in a cycle known as the *air-standard diesel cycle* shown in Figure P21.19. Fuel is sprayed into the cylinder at the point of maximum compression,  $B$ . Combustion occurs during the expansion  $B \rightarrow C$ , which is modeled as an isobaric process. Show that the efficiency of an engine operating in this idealized diesel cycle is

$$e = 1 - \frac{1}{\gamma} \left( \frac{T_D - T_A}{T_C - T_B} \right)$$



**Figure P21.19**

### Question 9      Freezer freezing Water

- 5.** A freezer has a coefficient of performance of 6.30. It is advertised as using electricity at a rate of 457 kWh/yr. (a) On average, how much energy does it use in a single day? (b) On average, how much energy does it remove from the refrigerator in a single day? (c) What maximum mass of water at 20.0°C could the freezer freeze in a single day? *Note:* One kilowatt-hour (kWh) is an amount of energy equal to running a 1-kW appliance for one hour.

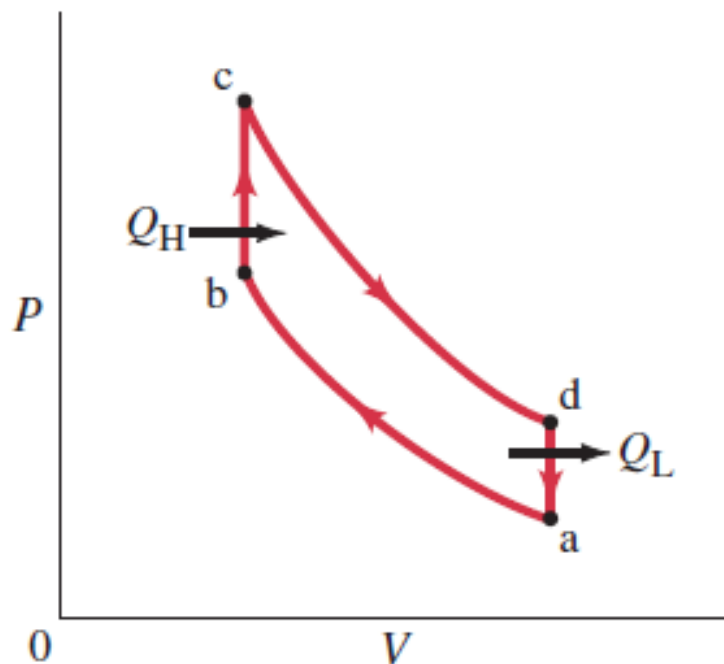


### Question 10      Heat Pump vs Air Conditioner

6. A heat pump has a coefficient of performance equal to 4.20 and requires a power of 1.75 kW to operate. (a) How much energy does the heat pump add to a home in one hour? (b) If the heat pump is reversed so that it acts as an air conditioner in the summer, what would be its coefficient of performance?

### Question 11      Compression Ratio of Otto Cycle

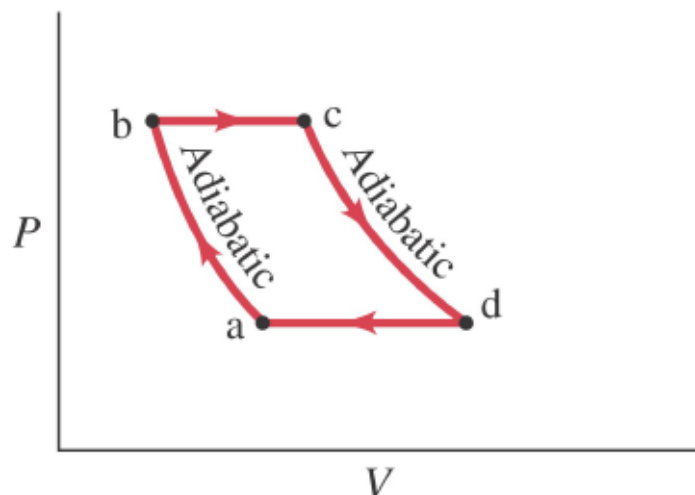
- \*21. (III) In an engine that approximates the Otto cycle (Fig. 8), gasoline vapor must be ignited at the end of the cylinder's adiabatic compression by the spark from a spark plug. The ignition temperature of 87-octane gasoline vapor is about  $430^{\circ}\text{C}$  and, assuming that the working gas is diatomic and enters the cylinder at  $25^{\circ}\text{C}$ , determine the maximum compression ratio of the engine.



**FIGURE 8** The Otto cycle.

## Question 12      Brayton Cycle

The  $PV$  diagram shown in Figure 5 is called the Brayton cycle which is made up of two adiabatic processes and two isobaric processes. Assume that ideal gas is being used in this cycle.



**Figure 5**

- (a) Assuming that we do not know if the ideal gas is monoatomic or diatomic, show that the efficiency of the Brayton cycle is  $e = 1 - \left(\frac{P_b}{P_a}\right)^{\frac{1-\gamma}{\gamma}}$ .
- (b) Suppose now we know that there are 2 moles of monoatomic ideal gas,  $P_a = 1$  atm,  $V_a = 0.05$  m<sup>3</sup>,  $P_b = 1.3$  atm and  $V_d = 0.07$  m<sup>3</sup>. Determine the change in internal energy  $\Delta U$ , heat exchange  $Q$  and work done by gas  $W$  for two of the processes in the Brayton cycle. Use the space below for your working neatly.

|                | $\Delta U$ | $Q$ | $W$ |
|----------------|------------|-----|-----|
| Process a to b |            |     |     |
| Process b to c |            |     |     |

- (c) Determine  $v_{rms}$  of the ideal gas at point d of the cycle.

Assuming the gas is Nitrogen with molar mass of 28 g.